



SUSTAINABLE OLYMPIAD

Sustainable Olympiad 2024

Round 1: Online Knowledge Challenge

Junior Division (ages 12-14)

Mark scheme and worked solutions

Total: 50 marks. Accept answers within rounding and correct alternative methods. Apply error carried forward (ECF): if a wrong value from an earlier part is used correctly in a later part, award the method marks.

Part A: Answer key

Question	Answer	Correct option
A1	B	0.1 kWh
A2	B	Stratosphere
A3	C	Wheat
A4	B	It increases, because water vapour diffuses away from the leaves faster
A5	B	36 t
A6	C	162 000 L
A7	A	Corals expel the algae living in their tissues when the water becomes too warm
A8	D	Glass bottle
A9	C	150 ha
A10	A	Gravitational potential → kinetic → electrical

Part A: Worked solutions

A1. Answer B

3 minutes = 0.05 h. Energy = 2.0 kW × 0.05 h = **0.1 kWh** (360 000 J).

A2. Answer B

About 90% of atmospheric ozone is in the **stratosphere**, roughly 15–35 km up, where it absorbs harmful UV radiation.

A3. Answer C

Wheat, like other grasses, is wind-pollinated (and mostly self-pollinating). Apples, almonds and strawberries need insects such as bees.

A4. Answer B

Heat, dry air and wind all steepen the water-vapour gradient between the leaf and the air, so transpiration **increases**. Stomata close only when the plant is short of water.

A5. Answer B

Last year: $0.30 \times 600 = 180$ t. This year: waste = 540 t, recycled = $0.40 \times 540 = 216$ t. Difference = **36 t**.

A6. Answer C

$30 \times 2 \times 2700 = \mathbf{162\ 000\ L}$, about the volume of a small swimming pool.

A7. Answer A

Corals depend on algae called zooxanthellae for food and colour. Heat stress makes the corals **expel the algae**, leaving the white skeleton visible. The coral may starve if it lasts too long.

A8. Answer D

Estimates: paper, weeks; aluminium can, about 200 years; plastic bottle, about 450 years; **glass**, up to a million years. Glass is inert, so it is far less harmful than plastic.

A9. Answer C

3 cm = 1.5 km and 2 cm = 1.0 km. Area = $1.5 \text{ km}^2 = \mathbf{150 \text{ ha}}$. Do not scale the area by 50 000; the area scale is $50\,000^2$.

A10. Answer A

Water stored high up has **gravitational potential** energy, which becomes kinetic energy as it falls, turning the turbine and generator to produce electrical energy.

Part B: Worked solutions and mark scheme

B1. A shorter shower (8 marks)

(a) [2 marks] Before: $8 \times 12 = 96$ L. After: $5 \times 8 = 40$ L. Saving = **56 L**.

Marking: 1 mark for both volumes; 1 mark for the saving.

(b) [2 marks] $56 \times 365 = \mathbf{20\,440\,L}$.

Marking: Allow error carried forward.

(c) [4 marks] Mass = 20 440 kg. Energy = $20\,440 \times 4.18 \times 25 = 2\,135\,980$ kJ.

Dividing by 3600 kJ per kWh gives **593 kWh**.

Marking: 1 mark for the mass; 1 mark for $\Delta T = 25\,^{\circ}\text{C}$; 1 mark for kJ; 1 mark for converting to kWh.

B2. Can trees cancel our emissions? (7 marks)

(a) [2 marks] $7500\text{ kg} / 22\text{ kg} = 340.9$, so **341 trees** (always round up).

Marking: 1 mark for converting tonnes to kg; 1 mark for the answer.

(b) [2 marks] $341 / 400 = \mathbf{0.85\,ha}$ (8500 m^2), larger than a football pitch.

Marking: Allow error carried forward.

(c) [3 marks] Any two well-explained points: young trees absorb very little for many years; the carbon is released again if the trees burn, die or are cut down; there is not enough land to offset everyone's emissions; cutting emissions is faster and permanent.

Marking: Up to 2 marks per reason (maximum 3).

B3. Counting daisies (7 marks)

(a) [1 mark] $50 \times 40 = \mathbf{2000\,m^2}$.

Marking: 1 mark.

(b) [2 marks] Total = 50 daisies in 10 m^2 , so the mean is **5.0 per m^2** .

Marking: 1 mark for the total; 1 mark for the mean.

(c) [2 marks] $5.0 \times 2000 = \mathbf{10\,000\,daisies}$.

Marking: Allow error carried forward.

(d) [2 marks] Random placement avoids bias, such as choosing spots with lots of daisies. More quadrats give a more reliable mean because daisies are not evenly spread.

Marking: 1 mark each.

B4. Electric or petrol? (8 marks)

(a) [2 marks] Electric: $16 \times 0.22 = \mathbf{€3.52}$. Petrol: $6.0 \times 1.80 = \mathbf{€10.80}$.

Marking: 1 mark each.

(b) [2 marks] Electric: $16 \times 0.25 = \mathbf{4.0\,kg}$. Petrol: $6.0 \times 2.3 = \mathbf{13.8\,kg}$.

Marking: 1 mark each.

(c) [2 marks] $12\,000\text{ km} = 120 \times 100\text{ km}$. Saving = $120 \times (10.80 - 3.52) = 120 \times 7.28 = \textbf{€873.60}$.

Marking: 1 mark for the number of 100 km units or the difference per 100 km; 1 mark for the answer.

(d) [2 marks] Any two of: making the battery causes large emissions; some energy is lost while charging; the grid's emissions vary with time and place; manufacturing the car itself.

Marking: 1 mark each.